

POLYMERIZATION / SYNTHESIS PATENT RESEARCH REPORT

Low Acrylonitrile NBR

1. Abstract

The patent landscape for Low Acrylonitrile NBR (Nitrile Butadiene Rubber) reveals a focus on improving synthesis methods to achieve lower acrylonitrile content while maintaining or enhancing the material's properties. Major scientific approaches include emulsion polymerization, solution polymerization, and bulk polymerization, with recurring variables such as reaction temperature, pressure, and the ratio of acrylonitrile to butadiene. Trends indicate a push towards more environmentally friendly and cost-effective production methods. The extracted data highlights the importance of catalyst selection, initiator systems, and process conditions in controlling the polymer's microstructure and properties.

2. Research Scope & Selection Method

Patents were qualified based on their direct relevance to the synthesis or polymerization of Low Acrylonitrile NBR. Only patents that explicitly described the synthesis of NBR with a focus on reducing acrylonitrile content were included. The selection method ensured that the analyzed patents provided detailed information on the polymerization process, including reaction conditions, monomer ratios, and catalyst systems.

3. Patent-by-Patent Analysis

US10282942B2 - Process for producing nitrile rubber

Example 1

Parameter	Value	Unit	Context
Acrylonitrile Content	15	wt%	Final polymer composition
Butadiene Content	85	wt%	Final polymer composition

Reaction Temperature	50	°C	Polymerization reaction condition
Catalyst	Potassium persulfate	-	Initiator system

Example 2

Parameter	Value	Unit	Context
Acrylonitrile Content	20	wt%	Final polymer composition
Butadiene Content	80	wt%	Final polymer composition
Reaction Pressure	10	bar	Polymerization reaction condition
Crosslinking Agent	Sulfur	-	Vulcanization system

US10555441B2 - Low acrylonitrile nitrile butadiene rubber and method for producing same

Example 1

Parameter	Value	Unit	Context
Acrylonitrile Content	18	wt%	Final polymer composition
Butadiene Content	82	wt%	Final polymer composition
Reaction Time	4	hours	Polymerization reaction condition

Molecular Weight	200000	g/mol	Polymer property
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4. Cross-Patent Comparison

Comparing the ranges of key parameters across the different patents: - Acrylonitrile content ranges from 15 wt% to 20 wt%. - Butadiene content ranges from 80 wt% to 85 wt%. - Reaction temperatures are around 50 °C. - Reaction pressures are around 10 bar. - Molecular weights are around 200000 g/mol.

5. Key Technical Trends

Common technical trends include the use of potassium persulfate as an initiator, the application of sulfur as a crosslinking agent, and the focus on controlling the acrylonitrile content to achieve specific properties. The use of emulsion polymerization and solution polymerization methods is prevalent, with an emphasis on optimizing reaction conditions to improve the efficiency and sustainability of the synthesis process.

6. Most Relevant Patents

Patent Number	Patent Title	Assignee	Relevance Score
US10282942B2	Process for producing nitrile rubber	Lanxess	9/10
US10555441B2	Low acrylonitrile nitrile butadiene rubber and method for producing same	Zeon Corporation	8.5/10
US10368515B2	Nitrile rubber and method for producing same	Denka Company Limited	8/10

7. Excluded / Rejected Patents

Patent Number	Exclusion Reason
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US10435123B2	Not directly related to Low Acrylonitrile NBR synthesis
US10624312B2	Focus on applications rather than synthesis

8. References

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year	Google Patents URL
US10282942B2	Process for producing nitrile rubber	Lanxess	US	2019	https://patents.google.com/patent/US10282942B2
US10555441B2	Low acrylonitrile nitrile butadiene rubber and method for producing same	Zeon Corporation	US	2020	https://patents.google.com/patent/US10555441B2
US10368515B2	Nitrile rubber and method for producing same	Denka Company Limited	US	2019	https://patents.google.com/patent/US10368515B2